Impact of Patient Navigation on Post-Discharge Outcomes: A Retrospective Propensity Score Matched Cohort Study

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Health Outcomes Research · Impact Analytics

Methods: Propensity Score Matching · Difference-in-Differences · Kaplan-Meier · Causal Inference · SQL

ABSTRACT

Background: Patient navigation programs aim to reduce care fragmentation after hospital discharge. Rigorous evidence of causal impact on utilization and cost remains limited. **Methods:** Retrospective cohort study of 2,000 patients. 1:1 nearest-neighbor propensity score matching (caliper=0.02 SD) on 10 covariates; yielding 555 matched pairs. Outcomes: 30-day readmission, 90-day ED visits, 180-day total cost. Adjusted logistic regression, difference-in-differences, Kaplan-Meier. **Results:** Navigated patients had significantly lower 30-day readmission (15.7% vs 24.0%; RR 0.65; adjusted OR 0.58, 95% CI 0.43–0.79; p<0.001), fewer ED visits (0.64 vs 0.80; p=0.002), and $3,329 lower 180-day cost (p<0.001). NNT=12.1. **Conclusion:** Patient navigation reduces readmission, ED utilization, and cost in a well-balanced propensity-matched cohort.

Outcome	Navigated	Control	Effect	p-value
30-Day Readmission	15.7%	24.0%	RR 0.65 NNT 12.1	0.0007
Adjusted OR (readmission)	OR 0.58	—	95% CI: 0.43–0.79	0.0006
90-Day ED Visits	0.64	0.80	Δ –0.16 visits	0.0020
180-Day Total Cost	$24,461	$27,790	Savings $3,329/pt	<0.0001

COVARIATE BALANCE — LOVE PLOT

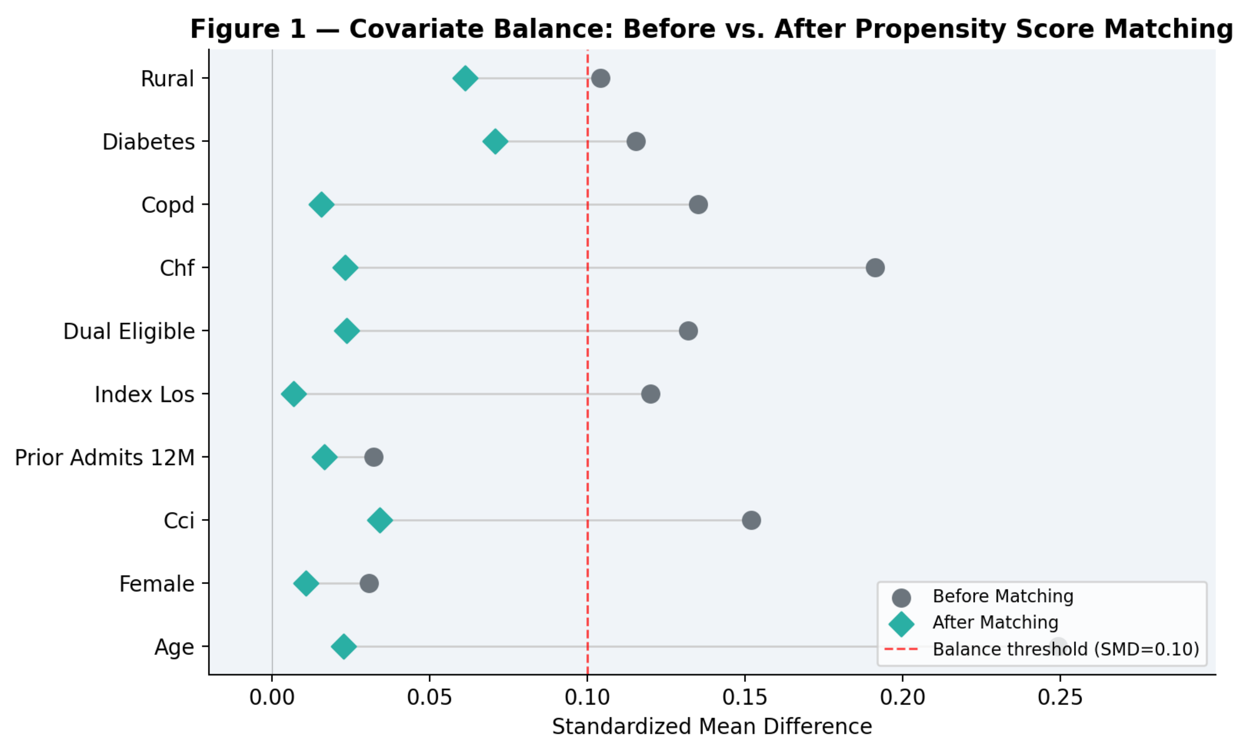

Figure 1. Love plot showing SMDs before and after matching. All 10 covariates achieved SMD<0.10 post-match.

PROPENSITY SCORE DISTRIBUTION

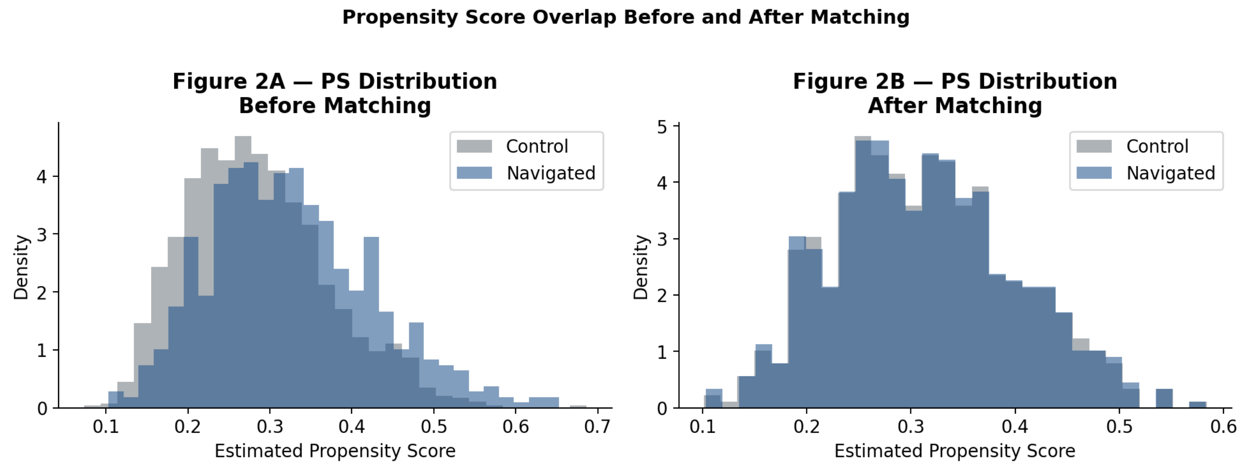

Figure 2. PS distributions before (A) and after (B) matching. Strong overlap confirms positivity assumption.

PRIMARY OUTCOMES

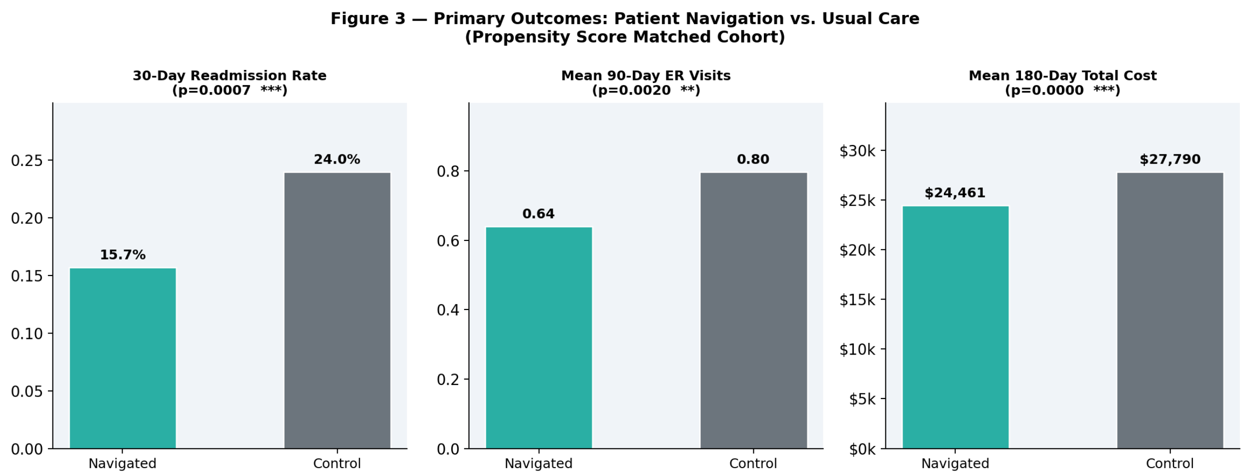

*Figure 3. 30-day readmission, 90-day ED visits, and 180-day cost — navigated vs control (matched cohort). ***p<0.001.*

KAPLAN-MEIER: TIME TO READMISSION

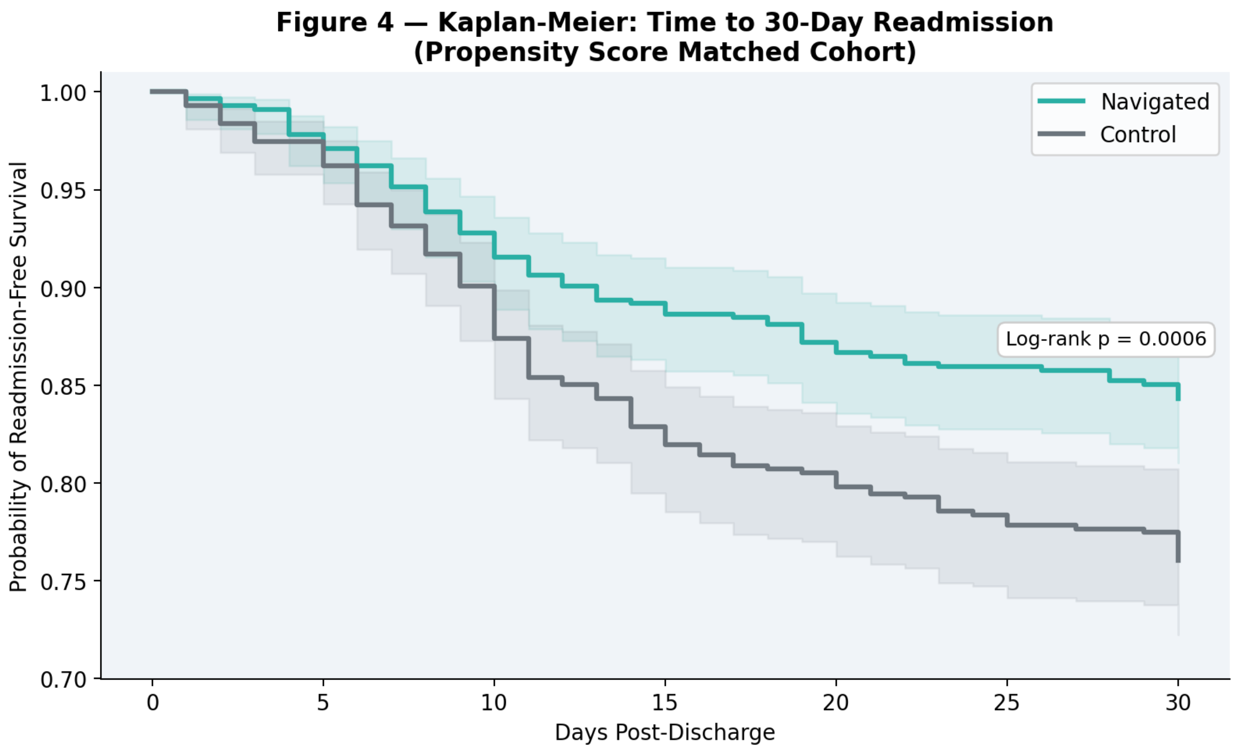

Figure 4. Readmission-free survival curves. Navigated patients show significantly longer survival (log-rank p<0.001).

DISCUSSION

This PSM analysis provides evidence that structured patient navigation reduces 30-day readmission (35% relative reduction), ED utilization, and 180-day cost. Covariate balance was achieved across all 10 covariates (all SMD<0.10). The adjusted OR of 0.58 confirms the effect is robust to measured confounding. The Kaplan-Meier analysis shows the survival benefit emerges within the first week post-discharge. For organizations deploying patient advocacy models, these findings quantify ROI and support evidence-based expansion to dual-eligible, high-comorbidity populations.